Traffic Data Collection and Analysis

Course Project - 3

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by

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Abstract

This study examines traffic operations at Sadhashivnagar PS Junction, Bengaluru, a signalized three-leg intersection, to understand traffic flow and signal performance. Manual surveys were conducted to collect data on traffic volumes, vehicle composition, queue lengths, spot speeds, and signal timings. The data were analyzed to identify traffic patterns, congestion points, operational challenges, and safety issues. The study also assessed the effectiveness of existing signal timing and intersection management strategies. The findings provide insights into improving traffic efficiency, safety, and user experience at urban signalized intersections.

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Chapter 1

Introduction

Traffic intersections are critical nodes within urban road networks, where multiple streams of vehicles converge and interact. The efficiency and safety of these intersections significantly influence overall traffic flow, congestion levels, and the experience of road users. Signalized intersections, in particular, require careful evaluation to understand traffic behavior, vehicle interactions, and the effectiveness of signal control systems.

Sadhashivnagar PS Junction, Bengaluru, is a signalized three-leg intersection with a heterogeneous mix of traffic, including two-wheelers, three-wheelers, autos, cars, light and heavy commercial vehicles, buses, and bicycles. The intersection experiences varying traffic volumes and movement patterns across its approaches, which affects operational performance, queue formation, and delays during peak hours.

This study focuses on collecting and analyzing key traffic parameters to assess intersection performance. The parameters include: traffic volumes to understand flow intensity and directional movement patterns; vehicle composition to identify the mix of different vehicle types; spot speeds to evaluate operating speeds under current traffic conditions; queue lengths to quantify congestion levels during signal cycles; and signal timings to assess phase duration, coordination, and overall signal efficiency.

The analysis of these data provides insights into the macroscopic traffic flow characteristics, operational efficiency, and safety at the intersection. Observations on signal performance, including green, amber, and red phase durations, highlight the effectiveness of traffic control strategies and their influence on queue formation and vehicular delays. Furthermore, the study considers pedestrian interactions, potential conflict points, and other operational conditions affecting user experience.

The findings from this study offer a detailed understanding of intersection operations, providing a technical basis for evaluating traffic management strategies, optimizing signal timings, and improving overall efficiency, safety, and mobility at urban signalized

intersections.

1.1 Objectives

The primary objectives of this study at Sadhashivnagar PS Junction are to:

1. **Analyze traffic flow patterns** by recording vehicle movements across all intersection approaches and understanding directional distribution.
2. **Assess vehicle composition** to identify the proportion of different vehicle types, including two-wheelers, three-wheelers, cars, buses, and commercial vehicles.
3. **Measure operating speeds** using spot speed studies to evaluate traffic movement efficiency and variations across vehicle classes.
4. **Quantify congestion levels** by measuring queue lengths during red signal phases and determining approach-wise average queue lengths.
5. **Evaluate signal performance** by recording the duration of green, amber, and red phases, determining cycle times, and analyzing the effectiveness of signal operations.
6. **Understand operational and safety conditions** by observing pedestrian interactions, conflicts, and other factors affecting intersection efficiency and user experience.
7. **Provide insights for traffic management** that can inform improvements in signal timing, intersection design, and overall traffic operations.

This study emphasizes systematic observation and analysis of intersection behavior to create a detailed understanding of traffic operations, which can serve as the basis for developing traffic simulation models and optimizing signalized intersection performance.

Chapter 2

Literature Review

Traffic data collection plays a crucial role in transportation planning, traffic management, and infrastructure design. Over time, data collection techniques have evolved significantly, shifting from manual surveys and fixed-point sensors to advanced, automated, and integrated systems. In India, this evolution has been shaped by rapid urbanization, mixed traffic conditions, and the need for cost-effective and scalable solutions.

Earlier studies and practices relied mainly on manual counting, pneumatic tubes, and inductive loop detectors, which provided accurate point-based measurements of flow, occupancy, and speed but suffered from limited spatial coverage, high maintenance requirements, and difficulties in long-term monitoring [1, 2]. These traditional methods were effective for small-scale or short-duration surveys but became inadequate for capturing continuous, network-wide data in dense urban environments.

To overcome these challenges, researchers and practitioners introduced Wireless Sensor Networks (WSNs) and Internet of Things (IoT)-based systems for traffic monitoring. These technologies enable automatic data transmission and remote accessibility, making them suitable for real-time applications. Studies have shown that WSNs can accurately measure vehicle counts and speeds while being more cost-efficient and easier to install than conventional loop detectors [3, 4]. However, in Indian conditions, the performance of WSNs can be affected by mixed traffic composition, lane indiscipline, and electromagnetic interference. Proper calibration, power supply management, and robust communication protocols are essential for maintaining accuracy in such environments.

At the network level, probe vehicle methods using GPS data have become increasingly popular in India. By equipping buses, taxis, and government fleet vehicles with GPS devices, continuous data on travel times, speeds, and congestion can be collected [5, 6]. Several studies in Chennai, Delhi, and Pune have demonstrated the effectiveness

of GPS-based systems in estimating corridor travel times and identifying congestion patterns. The use of floating car data (FCD) has also improved understanding of travel time variability and helped evaluate the performance of signalized corridors. However, accuracy depends on adequate penetration of probe vehicles and the consistency of data transmission.

To supplement probe data, Bluetooth and Wi-Fi sensors have been deployed to detect and match MAC addresses of passing devices between fixed detection points [7, 8, 9]. This approach helps estimate travel time and reliability over short links. In a study conducted in Chennai, Wi-Fi detection was successfully used to calculate travel time reliability. Nonetheless, such systems can be influenced by the rate of device detection, signal interference, and privacy-related issues such as MAC randomization. Despite these limitations, passive detection methods are widely appreciated for being cost-effective, portable, and less intrusive compared to conventional sensors.

In recent years, computer vision (CV) and artificial intelligence (AI) have become dominant technologies in traffic data collection and analysis [10, 11]. Many Indian cities, under the Smart Cities Mission, are installing AI-powered CCTV cameras, ANPR systems, and intelligent traffic management systems (ITMS). These systems use deep learning algorithms to automatically detect, classify, and count vehicles, identify traffic violations, and monitor congestion in real time. Studies in India show that such systems can identify a wide range of vehicle types, including two-wheelers, three-wheelers, and heavy vehicles, which are often missed by traditional detectors. However, challenges such as poor lighting, occlusion, dust, and non-standard license plate formats can reduce detection accuracy. To improve performance, algorithms must be trained on large and diverse datasets that represent Indian traffic conditions.

Highway authorities such as NHAI have begun deploying AI-based ATMS on corridors like the Dwarka Expressway to improve safety and enforcement [12, 13]. Similar deployments in cities such as Bengaluru, Mangaluru, and Madurai demonstrate that AI-based systems can reduce human intervention and improve monitoring efficiency. Yet, experts emphasize that these systems must comply with data privacy regulations and ensure transparency in algorithmic decision-making.

Another key trend observed in the literature is data fusion, which combines multiple data sources such as fixed sensors, probe vehicle data, and CV-based systems [14]. Fusion techniques improve accuracy by compensating for weaknesses in individual data types. For example, probe data can fill spatial gaps left by static sensors, while camera-based systems can validate probe-based estimates. Studies show that integrated systems provide more reliable and comprehensive information for estimating travel times, detecting incidents, and generating O–D matrices.

Indian researchers have also highlighted the importance of data quality control, including sensor health monitoring, calibration, and validation using ground-truth data. Continuous monitoring and retraining of machine learning models are recommended to handle changes in traffic flow patterns over time. Despite these advancements, challenges such as mixed traffic, varying road geometry, budget constraints, and institutional capacity remain critical barriers to large-scale implementation.

In conclusion, the literature indicates that India is transitioning towards a hybrid, technology-driven approach for traffic data collection. This involves using a combination of fixed detectors, wireless sensor networks, GPS probe data, and AI-enabled camera systems. Such integrated systems can deliver real-time, high-resolution data that supports congestion management, adaptive traffic control, and infrastructure planning [3, 5, 9, 14].

Chapter 3

Study Location and Data Description

3.1 Study Location

The study was conducted at Sadashivanagar Junction, a three-legged signalized intersection located in an urban area of Bengaluru. The intersection manages eastbound, westbound, and northbound traffic movements, with traffic signals controlling both vehicular and pedestrian flows to ensure safe and coordinated operation.

The junction accommodates a heterogeneous mix of vehicles, including two-wheelers, cars, buses, and light commercial vehicles, representing typical urban traffic conditions in the city. Traffic volumes are substantial throughout the day, with peak periods in the morning and evening leading to frequent signal cycles and queue formations across all approaches.

The surrounding land use is a combination of residential, institutional, and commercial establishments, contributing to steady traffic demand from all directions. A traffic control room is situated on the footpath opposite the northern approach, with traffic police monitoring both the signals and vehicular movement.

The study was conducted on Wednesday, 24th September 2025, between 6:30 PM and 8:00 PM under cloudy weather conditions. This location provides an appropriate setting for analyzing signalized intersection performance, traffic flow patterns, and operational efficiency under mixed traffic conditions typical of Indian urban road networks.



Figure 3.1: Location of Data Collection: Sadashiv Nagar Police Station

3.2 Data Collection and Methods

Traffic data were systematically collected to analyze intersection performance, including vehicular flow patterns, congestion, and signal operation. Standard manual data collection techniques were employed to capture traffic volumes, vehicle composition, queue lengths, and signal timings.

3.2.1 Classified Vehicle Volume Count

Traffic volumes were recorded for each approach, with vehicles classified into two-wheelers, three-wheelers, cars, light commercial vehicles (LCVs), heavy commercial vehicles (HCVs), buses, and bicycles. Turning movements were observed for all directions to understand flow distribution and approach-specific demand.

3.2.2 Vehicle Composition

Vehicle composition data were collected to quantify the proportion of each vehicle type. This information is essential for evaluating heterogeneous traffic behavior, as different vehicle classes affect intersection capacity, delays, and operational efficiency.

3.2.3 Queue Length Measurement

Queue lengths were measured during red signal phases. The extent of vehicles waiting at the signals was recorded until queues stabilized. Average queue lengths for each approach were calculated to quantify congestion and intersection performance during peak periods.

3.2.4 Signal Timing Measurement

Signal operations were observed, including the duration of green, amber, and red phases, as well as the overall cycle time for each approach. These measurements facilitate the assessment of signal efficiency, coordination, and their impact on vehicular delays.

3.2.5 Spot Speed Analysis

Spot speeds were measured along marked stretches of each approach. Both Time Mean Speed (TMS) and Space Mean Speed (SMS) were calculated to assess traffic flow characteristics:

- **Time Mean Speed (TMS)**: The arithmetic mean of vehicle speeds observed over a specific time period.

$$TMS = \frac{\sum_{i=1}^{n} v_i}{n}$$

  where v_i is the speed of the ith vehicle and n is the total number of vehicles observed.

- **Space Mean Speed (SMS)**: The harmonic mean of vehicle speeds measured over a defined road segment.

$$SMS = \frac{n}{\sum_{i=1}^{n} \frac{1}{v_i}}$$

  where v_i is the speed of the ith vehicle and n is the total number of vehicles observed.

3.2.6 Data Analysis

The collected data provide a comprehensive view of intersection performance, including traffic volumes, vehicle composition, speed variations, queue formation, and signal timing. This dataset forms the basis for analyzing traffic flow patterns, operational efficiency, and safety, and can be used for traffic simulation modeling in softwares.

Chapter 4

Collected Data Summary

4.1 Spot Speed Data Sheet

Location:	Sadhasivanagar PS Junction, Bengaluru	**Date & Day:**	24/09/2025, Wednesday
Name of Approach:	WB	**Weather:**	Cloudy
Time:	7:50 PM – 8:00 PM	**Distance:**	30 m

SI No.	2 Wheeler	3 Wheeler	Car	LCV	HCV	Bus	Bicycle	Others
1	10.00	10.00	15.00	5.00	7.50	10.00		10.00
2	10.00	5.00	6.00	7.50		6.00		6.00
3	7.50	10.00	10.00	5.00		7.50		7.50
4	10.00	7.50	10.00	5.00		3.33		15.00
5	10.00	5.00	10.00	5.00		3.33		15.00
6	10.00	6.00	7.50	6.00		4.29		
7	10.00	5.00	7.50	7.50		10.00		
8	15.00	7.50	10.00	5.00		6.00		
9	15.00	6.00	7.50	4.29		7.50		
10	7.50	7.50	10.00	5.00		7.50		
11	15.00	6.00	6.00	3.33		6.00		
12	15.00	6.00	6.00	3.33		7.50		
TMS	11.25	6.79	8.79	5.16	7.50	6.58		10.70
SMS	10.59	6.43	8.18	4.86	7.50	5.81		9.38

Table 4.1: Spot Speed Observations for Different Vehicle Types (WB Approach)

4.2 Volume Count Data Sheet

Location:	Sadhasivanagar PS Junction, Bengaluru	**Date & Day:**	24/09/2025, Wednesday
Time:	7:25 PM – 7:40 PM	**Weather:**	Cloudy

Movement	From	To	2W	3W	Car	LCV	HCV	Bus	Bicycle	Pedestrians	Others
Right Turn	North	West	99	24	70	2	1	6	0	2	1
Left Turn	North	East	24	10	21	2	0	0	0	1	0
Total	-	-	**123**	**34**	**91**	**4**	**1**	**6**	**0**	**3**	**1**

Table 4.2: Turning movement counts for Southbound Approach

Movement	From	To	2W	3W	Car	LCV	HCV	Bus	Bicycle	Pedestrians	Others
Through	East	West	397	58	180	33	7	8	0	3	0
Right Turn	East	North	202	44	101	5	0	0	2	6	1
Total	-	-	**599**	**102**	**281**	**38**	**7**	**8**	**2**	**9**	**1**

Table 4.3: Turning movement counts for Westbound Approach

Movement	From	To	2W	3W	Car	LCV	HCV	Bus	Bicycle	Pedestrians	Others
Through	West	East	270	52	147	17	5	6	3	1	1
Left Turn	West	North	69	22	49	1	0	3	0	2	0
Total	-	-	**339**	**74**	**196**	**18**	**5**	**9**	**3**	**3**	**1**

Table 4.4: Turning movement counts for Eastbound Approach

4.3 Intersection Diagram with Volume Count

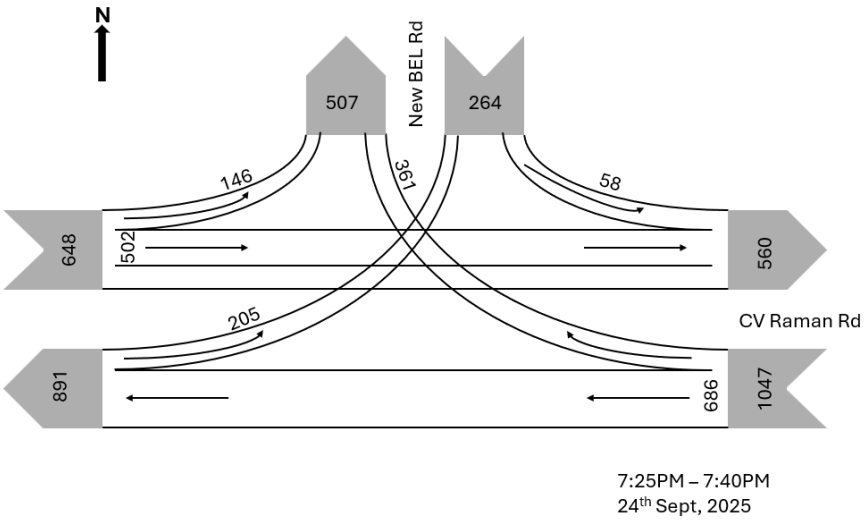


Figure 4.1: Intersection diagram of Sadhasivanagar PS Junction, Bengaluru

4.4 Queue Length Data Sheet

Location:	Sadhasivanagar PS Junction, Bengaluru	**Date & Day:**	24/09/2025, Wednesday
Time:	6:30 PM – 6:45 PM	**Weather:**	Cloudy

Eastbound	Westbound	Southbound
346.99m	354m	382m

Table 4.5: Queue Length observations

4.5 Traffic Signal Timing Data Sheet

Location:	Sadhasivanagar PS Junction, Bengaluru	**Date & Day:**	24/09/2025, Wednesday
Time:	6:57 PM – 07:12 PM	**Weather:**	Cloudy

Start of Green	Start of Amber	Start of Red	Red Light	Start of Amber	Start of Green	Cycle Length
↔ 00:00	00:10	← 00:13	02:37	05:14	↔ 05:17	—
↔ 05:17	06:20	← 06:22	08:07	10:33	↔ 10:36	05:19
↔ 10:36	12:48	← 12:51	15:30	17:07	↔ 17:10	06:36
↔ 17:10	18:21	← 18:24	20:15	21:12	↔ 21:14	04:04
↔ 21:14	23:28	← 23:31	25:26	27:09	↔ 27:11	05:57

Table 4.6: Signal Timing Data for SB Approach

Start of Green	Start of Amber	Start of Red	Red Light	Start of Amber	Start of Green	Cycle Length
–	–	00:00	00:13	02:30	←↑ 02:33	–
←↑ 02:33	05:15	← 05:17	06:22	10:31	←↑ 10:34	08:01
←↑ 10:34	10:53	← 10:55	15:13	15:16	←↑ 15:19	04:45
←↑ 15:19	18:15	← 18:18	18:55	19:15	←↑ 20:17	04:58
←↑ 20:17	22:51	← 22:32	23:32	25:22	←↑ 25:25	05:08

Table 4.7: Signal Timing Data for EB Approach

Start of Green	Green	Start of Amber	Start of Red Light	Start of Amber	Start of Green	Cycle Length
–	–	–	00:00	00:10	↑→ 00:13	—
↑→ 00:13	↑ 02:03	05:14	05:17	06:20	↑→ 06:23	04:58
↑→ 06:23	↑ 08:09	10:31	10:34	12:51	↑→ 12:54	06:28
↑→ 12:54	↑ 14:15	16:54	16:57	18:21	↑→ 18:24	05:30
↑→ 18:24	↑ 20:16	27:09	27:12	23:30	↑→ 23:33	05:09

Table 4.8: Signal Timing Data for WB Approach

Chapter 5

Traffic Movement Characteristics

The study intersection is a three-leg, signalized junction with eastbound (EB), westbound (WB), and southbound (SB) approaches catering to mixed urban traffic comprising two-wheelers, three-wheelers, cars, light and heavy commercial vehicles, buses, bicycles, and pedestrians. The surrounding land use is mixed residential, institutional, and commercial, generating steady demand throughout the day. During evening peaks, traffic police assist in regulating flow, yet opportunistic driving behavior and mixed-traffic dynamics result in frequent lane changes and stop-line encroachment.

Turning movement counts reveal a clear directional dominance of the westbound approach, which carries the highest through traffic and significant right-turn demand toward the northbound leg. The eastbound approach shows steady through movement with notable left-turns, while the southbound approach contributes moderate through volumes but heavy right-turn activity. This pattern establishes the E-W corridor as the major flow axis at the intersection. Because the westbound phase overlaps with multiple signal stages, it experiences nearly continuous vehicle movement, indicating high directional demand but also inefficient phase coordination that disadvantages the other legs.

Spot-speed studies on the westbound approach over a 30 m trap confirm low operating speeds typical of congested urban intersections. Two-wheelers recorded the highest mean speeds ($\approx$ 11 m/s), while LCVs and buses were the slowest ($\approx$ 5 - 6 m/s). The consistent difference between time mean and space mean speeds highlights the heterogeneous nature of flow and variable speed distribution near the stop line. The presence of heavy vehicles, combined with frequent lane changes, increases start-up lost time and delays complete queue clearance within a single green phase.

Average queue lengths of about 347 m (EB), 354 m (WB), and 382 m (SB) indicate severe congestion and residual queues that often spill back into upstream links. Illegal U-turns on the eastbound leg, lane indiscipline, and opportunistic pedestrian crossings

further reduce effective capacity and disrupt progression. Overall, the intersection exhibits high through demand on the E-W corridor, long red-phase queues, low speeds, and unbalanced signal timing. These factors underscore the need for optimized signal phasing and improved traffic discipline to enhance safety, reduce delays, and ensure smoother platoon movement across all approaches.

Chapter 6

Comparative Analysis

6.1 Spot Speed Analysis

The spot speed study was conducted on the Westbound approach of the Sadhashivanagar PS Junction over a 30 m trap length during the evening peak hour (7:50 PM – 8:00 PM) under cloudy weather conditions. The analysis of observed vehicle speeds reveals distinct variations among different vehicle categories, reflecting the heterogeneous nature of traffic flow at the intersection. The Time Mean Speed (TMS) ranged from 5.16 m/s (LCVs) to 11.25 m/s (two-wheelers), while the Space Mean Speed (SMS) varied between 4.86 m/s and 10.59 m/s across vehicle types. In all cases, TMS values were slightly higher than SMS, indicating speed variability and differences in acceleration among vehicles, typical of congested urban approaches.

Among all classes, two-wheelers recorded the highest average speeds (TMS = 11.25 m/s, SMS = 10.59 m/s), benefiting from greater maneuverability and the ability to utilize smaller gaps in traffic. Cars followed with moderate speeds (TMS = 8.79 m/s, SMS = 8.18 m/s), representing the general flow of the traffic stream. Three-wheelers and buses operated at relatively lower speeds ($\approx$ 6–7 m/s), while LCVs displayed the lowest average speeds (TMS = 5.16 m/s, SMS = 4.86 m/s), primarily due to slower acceleration and longer start-up times. The HCVs, though few in number, maintained consistent speeds around 7.5 m/s, constrained by size and limited maneuvering space. The observed difference between TMS and SMS across all categories confirms the heterogeneous composition and interference among vehicle classes, leading to non-uniform flow and reduced overall efficiency.

Speed Range (m/s)	Frequency	Middle Speed (m/s)	% Frequency	Cumulative % Frequency
3 – 5	15	4	22.73%	22.73%
5 – 7	12	6	18.18%	40.91%
7 – 9	16	8	24.24%	65.15%
9 – 11	16	10	24.24%	89.33%
11 – 13	0	12	0.00%	89.33%
13 – 15	7	14	10.61%	100%

Table 6.1: Speed Distribution Data

- The maximum frequency (24.24%) occurs in the 7–9 m/s and 9–11 m/s groups, indicating that most vehicles travel within this speed range.
- The cumulative frequency curve shows that about 89% of vehicles travel below 11 m/s, while only 11% exceed 13 m/s, suggesting that speeds above 13 m/s are rare.
- The mean (average) speed lies approximately around 8.5 to 9 m/s, which can be considered the representative speed for the flow under prevailing conditions.
- The mode (most frequent speed group) also corresponds to 7–9 m/s and 9–11 m/s, confirming a concentration of vehicle speeds in the mid-range.
- The median speed (corresponding to 50% cumulative frequency) occurs at about 8.5 m/s, which aligns closely with the mean — indicating a relatively symmetrical distribution of speeds.
- The absence of vehicles in the 11–13 m/s range creates a visible dip in the frequency curve, possibly due to temporary congestion, vehicle heterogeneity, or signal interruptions.
- The overall shape of the curves reflects mixed traffic conditions with moderate speeds, limited acceleration opportunities, and frequent interruptions from signal phases and pedestrian movements.

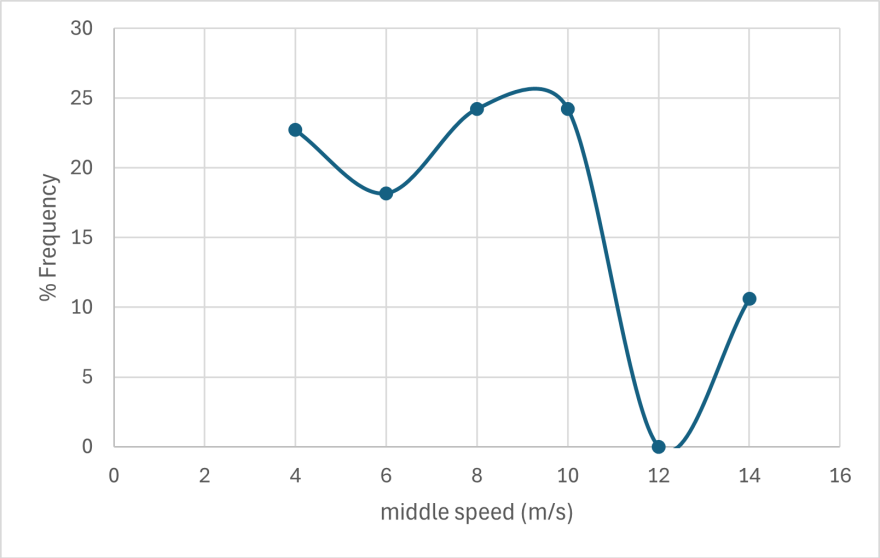


Figure 6.1: Variation of Percentage Frequency with Middle Speed for the WB Approach

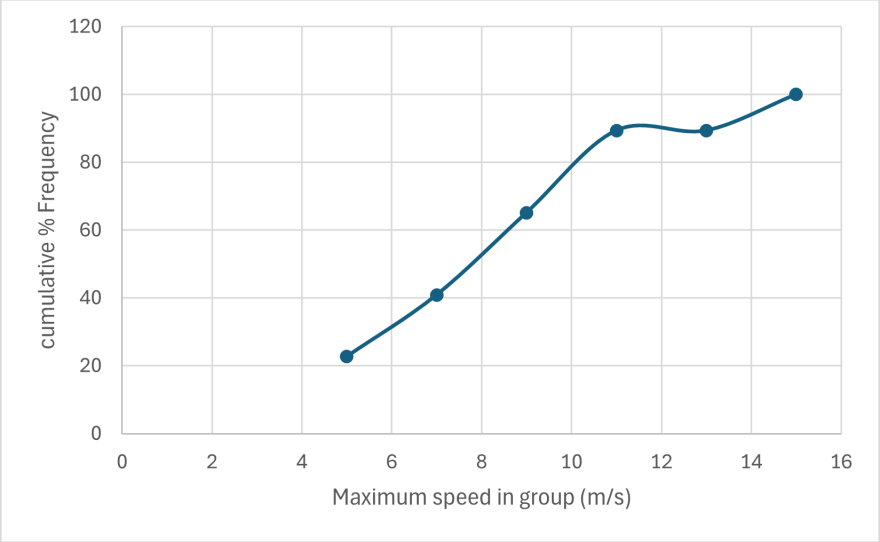


Figure 6.2: Cumulative Percentage Frequency Curve showing the progressive increase in vehicle proportion with speed for the WB Approach

6.2 Volume Count Analysis

The turning-movement counts reveal significant variation in approach-wise traffic demand at the Sadhashivanagar PS Junction. The westbound approach carries the highest total volume ($\approx$ 1,045 vehicles) during the 15-minute observation period, dominated by two-wheelers (599) and cars (281). This approach exhibits strong through movement toward the west and notable right-turn flow toward the northbound direction, indicating its role as the principal traffic corridor of the intersection. The eastbound

approach recorded a moderate total of about 647 vehicles, consisting mainly of through traffic toward the east (501) and some left-turn movement (146) toward the north. In contrast, the southbound approach reported the lowest total volume (264 vehicles), composed mainly of right-turning traffic from the north to the west. Despite its lower flow, the southbound approach often experiences longer delays, primarily because of shorter green allocation and its function as a minor leg feeding into the major E–W corridor.

Two-wheelers and cars together form the bulk of vehicular movement across all legs, reflecting the heterogeneous and mixed-traffic composition typical of Bengaluru's urban intersections. The high share of light vehicles promotes maneuverability but also increases lane indiscipline, particularly on the WB approach where overlapping signal phases cause near-continuous movement. The presence of LCVs, HCVs, and buses further lowers saturation flow rates and extends clearance time, leading to intermittent queuing even on the major legs.

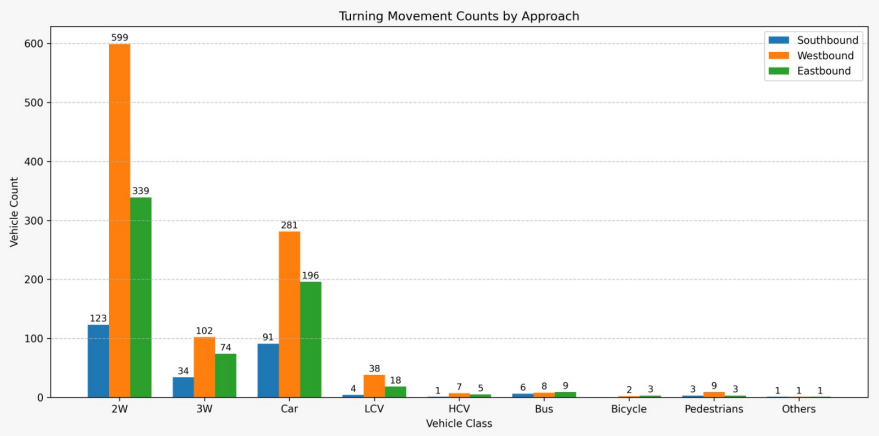


Figure 6.3: Number of vehicles of each class by Bound

6.3 Queue Length Analysis

Queue length observations were taken during the red phase for all three approaches to assess congestion levels and storage capacity utilization. The results show that the southbound approach experienced the longest average queue length ($\approx$ 382 m), followed by the westbound approach ($\approx$ 354 m) and the eastbound approach ($\approx$ 347 m). Despite having the lowest traffic volume, the southbound approach recorded the highest queue length, primarily due to longer red durations (about 190 seconds) and inadequate green splits in the existing fixed-time signal plan. The westbound approach, carrying the heaviest traffic volume, also exhibited substantial queues, reflecting that the current signal cycle is unable to efficiently discharge the accumulated vehicles during

each phase. The eastbound approach showed slightly shorter queues but still faced residual buildup, indicating overall oversaturation during the evening peak period.

Chapter 7

Observations on Signal Performance

Phase Duration and Cycle Characteristics

The average phase durations and total cycle times were computed for all approaches using the following standard equations:

- **Green Time (G):** Calculated as the difference between the start of amber and the start of green.

$$G = (StartofAmber - StartofGreen) \tag{7.1}$$

- **Amber Time (A):** Calculated as the difference between the start of red and the start of amber.

$$A = (StartofRed - StartofAmber) \tag{7.2}$$

- **Red Time (R):** Calculated as the difference between the start of the next green phase and the start of red.

$$R = (StartofNextGreen - StartofRed) \tag{7.3}$$

- **Cycle Length (C):** Calculated as the total duration of one complete signal cycle (green + amber + red).

$$C = G + A + R \tag{7.4}$$

These calculations were performed separately for each observed cycle to account for minor fluctuations in signal operation. Once individual cycle values were determined,

the **average phase duration** for each approach was computed to represent typical operational conditions. The average was calculated using the arithmetic mean formula:

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n} \tag{7.5}$$

where X_i represents the measured value of a phase duration (green, amber, red, or cycle length) in the ith cycle, and n is the total number of observed cycles. The resulting average values of green, amber, red, and total cycle times for each approach are presented in Table 7.1.

The computed average values for each movement are shown below:

Direction	**Green (s)**	**Amber (s)**	**Red (s)**	**Total Cycle (s)**
EB Through	152	3	249	404
WB Through	297	3	95	395
SB East	226	3	101	330
SB West	102	3	224	329
EB North	212	3	200	415
WB North	110	3	245	358

Table 7.1: Average Phase Durations and Cycle Times

The westbound approach receives the longest green time (∼297 s) due to high traffic volume, while the SB West movement receives the shortest (∼102 s), reflecting its lower demand. However, total cycle times (329–415 s) are significantly longer than recommended (120–160 s), leading to excessive waiting times and inefficient signal operation.

Signal Efficiency Analysis

Signal efficiency is a measure of how effectively the total cycle time is utilized for moving traffic (green time) versus time spent stopped (red + amber). It can be expressed as:

$$\eta = \frac{G}{C} \times 100 \tag{7.6}$$

where:

- η = signal efficiency (%)
- G = green time (s)
- C = total cycle time (s)

Key Findings:

Direction	Green (s)	Cycle (s)	Efficiency (%)
EB Through	152	404	37.6
WB Through	297	395	75.2
SB East	226	330	68.5
SB West	102	329	31.0
EB North	212	415	51.1
WB North	110	358	30.7

Table 7.2: Signal Efficiency by Approach

- The WB Through movement exhibits the highest signal efficiency (75.2%), indicating optimal green time allocation relative to cycle length.
- SB West and WB North movements show efficiencies below 32%, suggesting underutilized green phases and prolonged red times, resulting in excessive queuing.
- The overall signal efficiency ranges between 30% and 75%, below the desired 70–85% range for urban intersections, indicating scope for optimization.

Operational Insights

- **Oversized Cycle Length:** With average cycles of 330–415 s, the intersection suffers from excessive delay, especially during off-peak. Shortening cycles to 140–160 s would improve responsiveness.
- **Inefficient Phase Distribution:** Large discrepancies in efficiency (30–75%) indicate that green time is not aligned with actual demand.
- **Residual Queues:** Persistent queues show that even long greens are insufficient due to high demand and start-up lost times.
- **Pedestrian Phase Absence:** Lack of a dedicated pedestrian phase compromises safety, particularly on the WB approach.
- **Heterogeneous Traffic Effects:** Mixed vehicle types reduce saturation flow, increasing discharge time and lowering utilization efficiency.

The signal performance evaluation reveals that while the **WB approach** operates efficiently due to high green allocation and continuous discharge, other approaches, particularly **SB West** exhibit low efficiency, poor green utilization, and frequent residual queues. Overall, the intersection operates below its potential capacity due to imbalanced signal timing, excessive cycle length, and heterogeneous traffic flow.

Optimizing signal timings, redistributing green splits according to demand, and integrating pedestrian phases are essential to improving efficiency, reducing delays, and enhancing user safety at this intersection.

Chapter 8

Safety and Operational Conditions

Pedestrian Delays and Conflicts

- Long pedestrian wait times and insufficient crossing time during peak periods force pedestrians to wait excessively to cross.
- Pedestrians often cross during the green phase for vehicles, creating unsafe situations and increasing the risk of accidents.
- Pedestrian conflicts are evident as they tend to cross outside of designated crosswalks, especially when they perceive that the wait time is too long.

Congestion and Long Queue Lengths

- Queue lengths observed at the intersection indicate significant congestion, particularly in the Eastbound (346.99 m) and Westbound (354 m) directions.
- These long queues reflect that the signal cycle is not adequately handling the volume of traffic during peak hours, resulting in delayed vehicle movement and traffic bottlenecks.
- Congestion is exacerbated when vehicles are unable to clear the intersection during a single signal cycle, leading to spillback onto adjacent roads and increasing delays.

Illegal U-Turns and Vehicle Conflicts

- Illegal U-turns on the Eastbound approach cause traffic disruptions.

- These U-turns create temporary blockages, clogging up adjacent lanes and delaying traffic flow.
- Vehicles making illegal U-turns often end up in the middle of intersections or block adjacent lanes, which contributes to unnecessary traffic jams.
- Near-accidents were observed during these U-turns, where vehicles almost collided due to the blocking of lanes and abrupt changes in vehicle movement.

Minor Vehicle Collisions

- Five minor collisions were observed, where vehicles either tapped or scratched others, primarily due to the high traffic volume and lack of sufficient space between vehicles.
- These incidents were not severe but indicate a pattern of driver frustration and space management issues.
- The collisions occurred when vehicles tried to advance as much as possible during the green phase, sometimes pushing into already occupied spaces.
- This could be a result of tight lanes and drivers' behavior in trying to maximize their forward progress.

Traffic Police Exposure to Pollution

- Traffic police stationed at the intersection are exposed to high levels of air and noise pollution due to the continuous flow of vehicles and congestion.
- This exposure can have long-term health implications for the officers and impact their ability to perform their duties effectively.
- Officers themselves reported that the constant exposure to pollution is an issue, and they would appreciate improvements to reduce the effects of pollution in their work environment.

Vehicle Composition and Impact on Congestion

- The intersection handles a mix of vehicle types, including motorcycles, light commercial vehicles (LCVs), buses, and heavy commercial vehicles (HCVs).

- The diversity in vehicle size and speed characteristics can cause significant flow imbalances.
- Heavy vehicles (HCVs) and buses tend to slow down the traffic flow, especially during peak periods, leading to traffic jams.
- Motorcycles and smaller vehicles can cause blockages or move unpredictably when there is a lack of sufficient lane discipline, adding to congestion.

Inadequate Lane Discipline

- Vehicles are often seen changing lanes abruptly or cutting into other lanes, especially in congested conditions.
- This can disrupt the flow of traffic and increase the likelihood of minor collisions.
- Drivers seem to disregard lane markings in an attempt to gain positions during green light phases, leading to unpredictable vehicle movements and safety hazards.

Intersection Performance Evaluation

- In terms of safety, the intersection poses risks for both vehicles and pedestrians due to high vehicle volume and poor pedestrian facilities (e.g., long wait times).
- Efficiency is compromised due to extended queue lengths and bottlenecks that prevent smooth traffic flow.
- Vehicles are often unable to move through the intersection within a single green phase, leading to unnecessary delays and increased congestion.
- From a user experience perspective, drivers face frustration from delays, while pedestrians experience long waiting times and unsafe crossing conditions.
- The lack of adequate pedestrian facilities further exacerbates the negative user experience, with people crossing at inappropriate places or waiting too long to cross.

Signal Visibility and Compliance

- All signal heads were clearly visible from adequate distances on each approach, ensuring that drivers could anticipate phase changes and respond safely.
- The presence of a short amber interval (∼2–3 s) provided sufficient transition time for vehicles to either clear the intersection or stop safely before the red phase.

Queue Formation and Clearance

- Approaches with higher red durations (especially **SB**, ∼190 s) experienced longer queue lengths and slightly higher delays. However, queues cleared efficiently once the green phase was initiated, indicating that the allocated green time was adequate under observed traffic volumes.
- The EB approach, with the highest green allocation (∼281 s), operated smoothly with minimal queuing, reflecting effective operational conditions for that movement.

Traffic Flow and Coordination

- The overall intersection operated in a stable manner, with average cycle time ∼328.6 s and predictable phase progression.
- No significant incidents of red-light running or abrupt braking were observed, suggesting that drivers were generally compliant and signal timing supported safe operations.
- Minor fluctuations in cycle length indicate possible adaptive control or slight variations in detector response, which enhanced operational efficiency without compromising safety.

Chapter 9

Recommendations

Based on the traffic data collected and the field observations made at the Sadhasivanagar PS Junction, several recommendations are proposed to improve safety, efficiency, and overall operational performance of the intersection. The following suggestions are tailored to the existing geometric and spatial constraints of the site.

Intersection Operations and Traffic Flow

- Queue lengths in the eastbound and westbound approaches indicate that traffic demand during peak periods exceeds the effective discharge capacity. To alleviate congestion, better enforcement of lane discipline and minimization of unnecessary lateral movements are recommended.
- Minor geometric refinements, such as clear demarcation of stop lines and repainting of lane markings, can help guide vehicles into more orderly queues and reduce spillback effects.
- Introduction of advanced direction markings on the pavement can improve approach alignment and reduce hesitation or late maneuvers near the intersection.
- Vehicles waiting to turn right or left should be encouraged to position themselves properly within the available lanes through pavement arrows and signage, preventing blockage of through-traffic.

Pedestrian Facilities and Safety

- Pedestrian crossing phases should be reviewed to ensure adequate crossing time, especially during evening hours when volumes are higher and visibility is reduced.

- Pedestrian push-button signals or a short all-red interval allowing pedestrians to cross safely could be explored if warranted by observed pedestrian demand.
- Zebra crossings should be repainted and maintained with high-contrast thermoplastic paint to ensure visibility, especially under low-light conditions.
- Rather than installing rigid barriers, the intersection design should follow the principles of *forgiving infrastructure* — using measures such as refuge islands or curb extensions to guide pedestrians naturally without creating harsh separation from the roadway.
- Improving street lighting at pedestrian crossing points will further enhance safety and comfort.

Traffic Management and Enforcement

- Enforcement against illegal U-turns should be strengthened, as such maneuvers currently cause localized disruptions and conflicts. Clear "No U-Turn" signage and painted directional arrows can reinforce movement discipline.
- On-site observation indicated frequent lane changes and encroachment beyond stop lines. Deployment of temporary cones or reflective studs can be considered during peak hours to channelize traffic.
- Regular monitoring by traffic police should continue during evening peaks to manage flow and respond to conflicts promptly.
- Awareness campaigns targeting two-wheeler riders and auto-rickshaw drivers can improve compliance with lane discipline and stopping behavior.

Environmental and Operational Improvements

- Prolonged idling at red signals contributes to localized air and noise pollution. Awareness messages or signboards promoting engine idling reduction can help mitigate emissions.
- Considering the high exposure of traffic police to pollutants, provision of shaded or ventilated booths at strategic locations near the intersection is recommended.
- Regular cleaning and maintenance of the intersection area can enhance visibility, improve safety, and foster better driver compliance.

Signal Timing Optimization

- The existing average cycle (∼329 s) is significantly longer than ideal and leads to unnecessary delays and longer queues.
- It is recommended to reduce the cycle length to around 140–160 s during peak hoursand 90–120 s during off-peak periods, depending on observed demand.
- Shorter cycles improve vehicle progression, reduce average control delay, and minimize pedestrian waiting time.

Approach	**Green Time (s)**	**Amber (s)**	**All-Red (s)**	**Total (s)**
WB (Major)	77	4	2	83
EB (Moderate)	43	4	2	49
SB (Minor)	15	3	2	20
Total Cycle	–	–	–	**150 s**

Table 9.1: Suggested Signal Timing Plans: Peak Hour (Approx. 150 s)

Approach	**Green Time (s)**	**Amber (s)**	**All-Red (s)**	**Total (s)**
WB (Major)	49	3	2	54
EB (Moderate)	27	3	2	32
SB (Minor)	10	3	1	14
Total Cycle	–	–	–	**100 s**

Table 9.2: Suggested Signal Timing Plans: Off-Peak Hour (Approx. 100 s)

Data Collection and Future Study

- Additional data collection during morning and weekend peak hours is suggested to obtain a more representative understanding of traffic behavior.
- Recording saturation flow rates and pedestrian crossing delays would help in future signal optimization studies.
- Integration of the collected data into a base simulation model in PTV Vissim can help evaluate alternative control strategies, geometric configurations, and pedestrian crossing schemes before real-world implementation.

Summary of Expected Benefits

Implementation of these recommendations is expected to:

- Reduce congestion and improve throughput during peak periods.
- Enhance pedestrian safety through improved markings and crossing time.
- Decrease instances of unsafe U-turns and lane encroachments.
- Provide a cleaner and safer working environment for on-duty personnel.
- Serve as a more reliable data foundation for future intersection performance modeling.

Chapter 10

Conclusion

The study at Sadhasivanagar PS Junction provided a clear understanding of the operational conditions at a signalized urban intersection. Field data on classified volumes, spot speeds, queue lengths, and pedestrian activity highlighted key issues such as congestion on the eastbound and westbound approaches, inconsistent lane discipline, and limited pedestrian safety provisions.

Although the intersection faces spatial constraints that limit major geometric changes, several context-appropriate measures—such as improved markings, enforcement of movement discipline, and better pedestrian guidance—can substantially enhance safety and efficiency.

The data collected will also serve as a valuable input for developing a base simulation model in PTV Vissim, enabling future evaluation of alternative control and management strategies. Overall, the exercise emphasizes that careful data collection and observation form the foundation for effective, evidence-based traffic management decisions.

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